Identifying and Analyzing Broadband Internet Reverse DNS Names

Youndo Lee
University of Maryland
ydlee@cs.umd.edu

Neil Spring
University of Maryland
nspring@cs.umd.edu

ABSTRACT

Reverse DNS (rDNS) names often contain information not only about core routers in some networks but about hosts on edge networks. Although the rDNS names of routers have been decoded and their naming patterns verified by researchers, the information extracted from edge host rDNS names has been limited to intuitive keywords. In this paper, we develop a methodology for identifying and analyzing rDNS names for broadband Internet end hosts without using any pre-defined keyword. With this methodology, we study whether ISPs use intuitive keywords in their rDNS names and whether rDNS names embed correct information (e.g., whether keyword "cable" indicates cable clients but not fiber or DSL clients).

CCS CONCEPTS

• **Networks** → **Network measurement**; **Naming and addressing**;

ACM Reference Format:
Youndo Lee and Neil Spring. 2017. Identifying and Analyzing Broadband Internet Reverse DNS Names. In *Proceedings of CoNEXT '17*. Incheon, Korea, 6 pages. https://doi.org/10.1145/3143361.3143392

1 INTRODUCTION

Reverse DNS (rDNS) names can expose information that ISPs encode about Internet hosts. Information extraction from rDNS names, however, is a challenging problem mainly because most ISPs have unique naming conventions. An initial work [1] on decoding rDNS names have used manual inspection to infer router locations and generated hand-crafted rule sets. Recently, machine-learning-based approaches have been proposed to infer the structure of rDNS names [2] and geographical locations [3]. These techniques commonly focus on analyzing the rDNS names of routers to support research on the Internet's core, such as topology discovery.

RDNS names can also provide useful information for research on edge networks. For example, rDNS names can help identify broadband client addresses to use as probe destinations in studies [4–6] and to characterize measurement results [7, 8]. Unlike decoding of rDNS names of routers, a common practice of decoding rDNS names for identifying broadband client addresses is to check whether rDNS names contain pre-defined keywords indicating broadband clients such as client, adsl, cable, ppp, res and dhcp [4–7]. This approach is simple to implement but is based on the expectation that rDNS names of broadband client addresses would have common keywords that can be enumerated. However, ISPs typically have their own rDNS naming conventions despite efforts for standardization [9].

In this paper, we develop a technique that identifies rDNS names for client addresses without using any pre-defined keyword. We use clustering to find similar rDNS names and identify clusters of rDNS names consisting of broadband client rDNS names by using traceroute results. We then distinguish between clusters for clients with different connection types such as DSL, cable, fiber and mobile connections by using latency characteristics of different access technologies. The existing techniques to decoding rDNS names of routers may be used for distinguishing between rDNS names for routers and clients but not for distinguishing between rDNS names for different connection types.

Our technique is useful for identifying client addresses and their connection types by rDNS names. The advantage of using rDNS names for host identification is that it does not incur measurement loads (given the publicly available rDNS dataset [10]) and can apply to non-responding addresses. The ability to identify hosts by rDNS names can help researchers who study broadband networks or the Internet without privileged data from ISPs. It can also help Internet companies, e.g., CDN providers that want to redirect clients with different connection types to different front-end servers that implement different optimization techniques.

The media type inference is also useful for investigating whether ISPs maintain rDNS names. Many ISPs have been upgrading or extending their broadband Internet services, e.g., from DSL/cable to fiber [11]. Mobile services also have been expanding [12]. If ISPs do not update rDNS names as they upgrade their access technologies, their rDNS names may contain incorrect information. For example, the rDNS names of fiber clients may have the string "dsl" or "cable" which is obviously incorrect. We apply the technique to 103 broadband ISPs and analyze whether ISPs embed correct information in rDNS names.

The remainder of the paper is structured as follows. We describe and validate the clustering technique that identifies rDNS names for clients in Section 2 and the media type inference technique that distinguishes between names for different connection types in Section 3. Section 4 analyzes results of the applications of our technique to 103 ISPs. We describe the related work in Section 5 and conclude with future work in Section 6.

CoNEXT '17, December 12–15, 2017, Incheon, Korea

ACM ISBN 978-1-4503-5422-6/17/12...$15.00
https://doi.org/10.1145/3143361.3143392

2 DETECTING CLIENT RDNS NAMES

2.1 Methodology

We do not use any pre-defined keyword to identify rDNS names for broadband client addresses. We analyze the entire rDNS names that each ISP has. We use clustering to aggregate similar rDNS names and analyze each cluster to identify clusters consisting of client addresses. We first describe clustering and then classification of clusters.

2.1.1 Generating clusters. ISPs typically have their own naming conventions and therefore we cluster rDNS names of each ISP independently. For each ISP,[1] we first obtain IPv4 address ranges allocated to the ISP using MaxMind GeoLite ASN databases [13]. We then choose two IPv4 addresses from each /24 block[2] within the obtained address ranges and perform a reverse DNS lookup for each chosen address.

We cluster the rDNS names by using hierarchical agglomerative clustering [16]. We select hierarchical clustering because it supports any pair-wise distance and has an advantage that the number of clusters does not have to be given in advance. Hierarchical clustering requires a matrix of similarity between every pair of entities (here, rDNS names). To estimate the similarity between the rDNS name pairs, we compute Levenshtein distance (i.e., edit distance) [17] between all the rDNS name pairs. When computing edit distances, we replace IP addresses in the rDNS names as a common keyword *{ipaddr}*.[3] This helps to prevent two client addresses with the same rDNS pattern but very different IP addresses from having low degree of similarity. We use an R library for hierarchical clustering and select the complete linkage method [18].

2.1.2 Data cleaning. Mapping AS to IPv4 address range is straightforward but mapping ISP name to AS is not trivial and can be erroneous. The issue is that different ISPs can have the same name and thus irrelevant ASes may be involved. To address this issue, we confirm whether the mapped ISPs are broadband ISPs using CAIDA's AS to organization mapping dataset [19] that contains more information about ASes such as country names and AS types. We also inspect the suffixes of rDNS names of the mapped ASes to be certain that all the ASes belong to the ISP.

Although we replace IP addresses in rDNS names with a common string, rDNS names of some ISPs can still have numbers specific to individual IP addresses. For example, some ISPs embed only the last two octets of IP addresses and some others write the last three octets first and then the first octet. We manually inspect rDNS names to identify the patterns of address-specific numbers, and then replace the numbers matching the patterns with a common string. The manual inspection is to go through clustering results rather than individual rDNS names. Since the number of broadband ISPs is manageable [11] and their rDNS name patterns are unlikely to change, the cost of the manual inspection will not be high.

[1]We describe how we obtain a list of broadband ISPs in Section 4.

[2]/24 blocks are highly correlated with rDNS names and traceroute [14, 15]. We select one address from each /25.

[3]If the octets of IP addresses are written in reverse order, we instead use the keyword *{ipaddr_rev}*.

2.1.3 Finding clusters for clients. We determine whether clusters consist of rDNS names for clients based on the size and purity of clusters.

Broadband ISPs are likely to allocate the majority of their IP addresses to client hosts in that a single access router serves many client hosts [20]. We thus expect that broadband ISPs would have dominant rDNS names. We have inspected clusters and actually found that all but one of ISPs[4] have 1-3 clusters that account for more than 70% of the total rDNS names that ISPs have. We tentatively determine such large clusters to be clusters for clients.

We next evaluate the potential clusters by finding patterns that only match the rDNS names within the clusters and checking if the rDNS names of routers match the patterns. To identify routers, we perform traceroute to one IP address for each /24 within the IP address ranges of each target ISP. We select addresses appearing in traceroute except the source and destination because they are obviously router addresses. If the patterns found from the clusters match at most T% of the rDNS names of the router addresses, we consider that the patterns of rDNS names in the clusters indicate clients. T can be viewed as the confidence of the rDNS patterns, in that T will correspond to the false positive rate of the classification between clients and routers by the rDNS patterns. To achieve a 95% confidence level, we use 5% for T.

2.2 Validation

In this section, we validate our methodology. Specifically, we verify whether the patterns found by our methodology accurately classify the addresses of clients and routers.

2.2.1 Collecting validation data. We use a Bitcoin nodes crawler called Bitnodes [21] to collect broadband client addresses. Bitcoin nodes are likely to be end hosts and thus, if nodes are in broadband access networks (i.e., if the addresses of nodes belong to broadband ISPs), we select them as broadband client addresses. Router addresses are straightforward to obtain. We perform traceroute towards all the collected client addresses and select hosts belonging to broadband ISPs except the sources and destinations. Note that this set of router addresses are distinct from the set we used for measuring the purity of the potential clusters in Section 2.1.3.

We validate our methodology for 8 broadband ISPs for which we could obtain at least 60 client and router addresses using Bitnodes and traceroute. Table 1 shows a list of the ISPs[5] along with the numbers of client and router addresses we collected for validation.

2.2.2 Finding rDNS patterns. We applied our methodology to the 8 ISPs and found rDNS patterns for client addresses in 6 ISPs. Table 2 shows the patterns we found. The patterns are diverse in terms of both structure and keywords. Whereas BT Group and Time Warner have the patterns ending in keywords, the others have the patterns beginning with keywords. Keywords are also very different. Time Warner uses the keyword "res" which can be considered intuitive. Verizon also has the intuitive keyword "pool" but another keyword "static" is not intuitive for client addresses in

[4]We describe ISPs that we have inspected in Section 2.2 and 4.

[5]Although Time Warner cable was acquired by Comcast, we regard them as separate ISPs because the rDNS naming schemes of Timer Warner cable did not change. For the same reason, we separate Virgin Media from Liberty Global who acquired Virgin Media in 2013.

ISP	Service Region	Clients	Routers
BT Group	UK	77	107
Comcast	USA	565	986
Cox	USA	263	145
Liberty Global	International	166	205
Shaw Comm.	Canada	61	91
Time Warner	USA	230	384
Verizon	USA	274	61
Virgin Media	UK	121	134

Table 1: The list of the ISPs selected for validation. The number of client and router addresses collected for each ISP is also shown.

ISP	Pattern in Regular Expression
BT Group	(btcentralplus | btopenworld).com$
Comcast	^c-.+comcast.net$
Shaw Comm.	^s010.+shawcable.net$
Time Warner	res.rr.com$
Verizon	^(pool | static).+verizon.net$
Virgin Media	^(cur | spc | cpc).+virgin

Table 2: The rDNS name patterns for client addresses. Cox and Liberty Global do not have dominant patterns that match at most 5% of the routers.

that routers are likely to have static addresses. The other ISPs have non-intuitive keywords which are unique to them. These results suggest that it may not be reasonable to identify client addresses by using pre-defined keywords.

Cox appears not to keep rDNS names up-to-date. We could find a large cluster containing about 70% of the rDNS names whose pattern is ^ip.+cox.net$. However, the pattern matched about 30% of the rDNS names of routers. This might be caused by Cox reassigning addresses for clients to routers without updating their rDNS names. For the same reason, we could not analyze Liberty Global either.

2.2.3 Evaluating rDNS patterns. We now measure the quality of classification by the patterns.[6] We use the true positive rate (TPR) and true negative rate (TNR) as metrics. Specifically, TPR is the number of the client addresses that match the pattern divided by the total number of client addresses in the validation data. TNR is calculated as the number of the router addresses that do not match the pattern divided by the total number of router addresses in the validation data. Table 3 shows the TPR and TNR of the patterns in percentages.

All the patterns achieve high TPRs. The TPR is at least 97% for the patterns of 5 ISPs and 3 of them have the 100% TPR. Comcast has the relatively low TPR. The TNR is also generally high. 5 ISPs

[6]Note that we evaluate only 6 (of 8 ISPs) for which we could find patterns. Hence, our analysis in this section may not represent the entire broadband ISPs.

	BT Group	Comcast	Shaw Comm.	Time Warner	Verizon	Virgin Media
TPR	100.00	93.62	98.36	100.00	100.00	97.52
TNR	98.13	97.05	100.00	94.27	100.00	100.00

Table 3: The true positive rate (TPR) and true negative rate (TNR) of the patterns.

have at least 97% TNRs. The TNR for Time Warner is relatively low as 94.27% but the pattern we have found corresponds to the published rDNS naming conventions of Time Warner [22]. These results show that our methodology successfully finds rDNS naming patterns for client addresses.

3 ANALYZING CONNECTION TYPES

Many ISPs maintain heterogeneous broadband access networks such as DSL, cable, fiber and mobile access networks. The clusters we identify in Section 2.1.3 may consist of rDNS names for clients with a specific connection type. In this section, we describe the media type inference technique that determines whether each cluster indicates a specific connection type. We first describe a method that identifies clusters for mobile connections and then methods that differentiate clusters for fiber connections from those for DSL/cable connections.

3.1 Identifying mobile connections

3.1.1 Methodology. We identify clusters consisting of rDNS names for clients with mobile connections by using a distinct characteristic of mobile links. A recent study on timeouts has observed that, if the RTT of an initial probe to a certain address is higher than that of subsequent probes, the address is likely cellular (i.e., mobile) [23]. Based on this observation, we use the difference between RTTs of an initial and subsequent pings as an indicator of mobile connections. Specifically, for each address associated with a rDNS name in a cluster, we send 10 pings once a second and then compute the difference between the RTT of the first ping and the minimum RTT of the rest. We repeat this 10 times with an interval of 60 seconds and take the maximum of the difference values. If the maximum difference is greater than D ms for at least R% of the addresses in the cluster, we determine the cluster to be mobile (i.e., the rDNS names in the cluster indicate mobile clients).

3.1.2 Calibration. In order to justify the method and determine the thresholds D and R, we measure the RTT differences on the sets of mobile and fixed client addresses. The client addresses that we have collected by using Bitnodes (Section 2.2.1) are all likely fixed client addresses in that mobile devices are unlikely to participate in Bitcoin. We use these addresses for calibration. We collect mobile client addresses by using Hobbit blocks that are the aggregates of addresses having common last-hop routers [14]. Recent studies have shown that cellular carriers connect their mobile networks with the Internet through a few ingress points. [24, 25]. This means that many mobile devices will share a common ingress point that can be seen as a last-hop router on the Internet topology. Hence, if ISPs that provide only mobile services (e.g., Verizon Wireless) have large Hobbit blocks, then addresses in the blocks are all likely mobile client addresses. Verizon Wireless (AS 22394) and Claro S/A

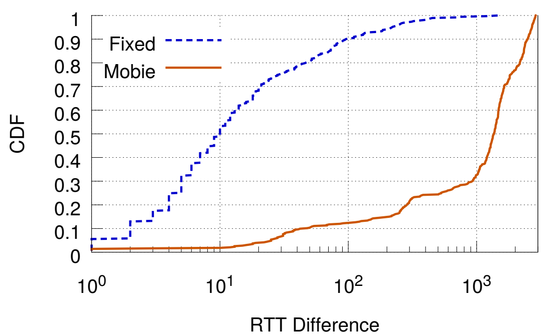

Figure 1: The difference between the RTT of the first ping and the minimum RTT of the rest for fixed and mobile broadband clients.

(AS 22085) had Hobbit blocks containing at least 100 /24 prefixes and we use the addresses in those blocks for calibration.

Figure 1 shows the CDFs of the RTT differences of fixed and mobile addresses. Fixed and mobile addresses have clearly different RTT differences. About 83% of the fixed and 11% of the mobile addresses have at most 60ms RTT differences. This result shows that the RTT difference is an effective indicator for mobile and fixed addresses. We use this result for determining the thresholds D and R. We aim to choose D such that best classifies the mobile and fixed addresses. Thus we find the RTT difference that maximizes information gain as in decision trees [26]. The information gain is largest when the RTT difference is 240.5ms[7] and about 84% and 4% of the mobile and fixed addresses have RTT differences greater than 240.5ms. Based on this result, if the RTT difference is greater than 240.5ms for at least 84% of the addresses in the cluster, we determine that the cluster consists of mobile client addresses. Similarly, if at most 4% of the addresses in the cluster have the RTT differences greater than 240.5, we determine that the cluster consists of fixed client addresses. Otherwise, we consider that the cluster contains both mobile and fixed addresses.

3.2 Identifying fiber connections

3.2.1 Methodology. Many ISPs have been upgrading their fixed broadband services from DSL/cable to fiber [11]. This often results in the coexistence of clients with fiber and DSL/cable connections. Once we determine whether clusters are for clients with mobile or fixed connections (Section 3.1), we differentiate clusters for fiber connections from those for DSL/cable connections.

Clients with fiber connections generally have lower latency than those having DSL/cable connections [27]. Fiber clients are also likely to have lower variance in latency in that both DSL and cable clients tend to have high variance [28]. Based on these observations, we use last hop latency[8] and the variance of latency for identifying clusters consisting of fiber clients. Specifically, for each address (associated with a rDNS name) in a cluster, we send 20 pings and compute the variance of RTTs. We estimate last hop latency by identifying a last-hop router of a target address using traceroute and subtracting the RTT to the last-hop router from that to the target address. We measure the last hop latency 20 times and use the median value. We determine that the rDNS names in the cluster are used for fiber clients if at least F% of the addresses have the last hop latency or variance[9] smaller than L.

3.2.2 Calibration. We collect data for calibration from a publicly available list of RIPE atlas probes [29] instead of using Bitnodes.[10] The owners of the probes may provide information about their Internet connection through user tags. Using this information, we identify the addresses of 494 fiber, 612 cable and 535 DSL broadband clients. We measure the variance of RTTs and last hop latency on these addresses. Figure 2 shows the results. The variances for cable clients tend to be larger than those for fiber clients. The cable clients even have larger variances than DSL clients. This could be because cable is a shared medium whereas DSL clients have dedicated lines [28]. The variances for DSL and fiber clients are similar. This result suggests that the variance of RTTs is not a good criterion for distinguishing between DSL and fiber clients. However, the last hop latency for DSL and fiber clients are very different. Two-thirds of the DSL clients have at least 10ms last hop latency whereas only about 6% of the fiber clients have the latency greater than or equal to 10ms. The latency for cable and fiber clients are also largely different but the variance of RTTs is a

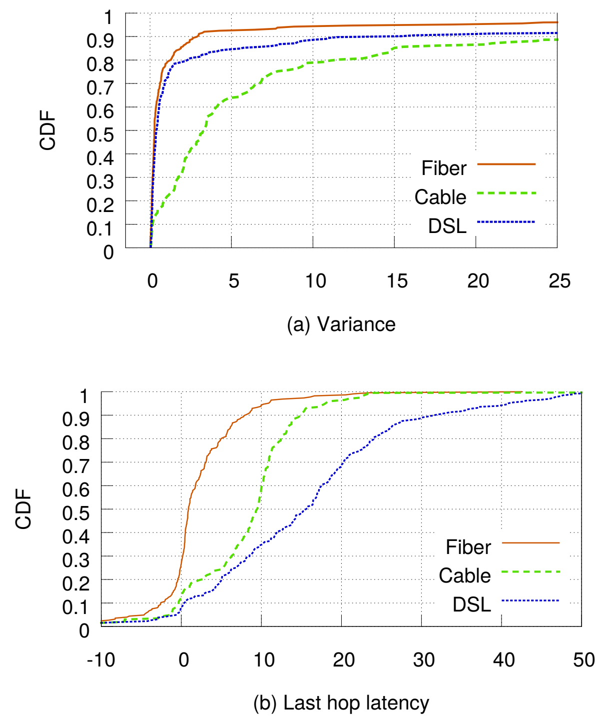

Figure 2: The CDFs of a) the variances of RTTs and b) the last hop latency of fiber, cable and DSL clients.

[7]Note that we do not use the threshold value for classifying individual addresses. We only use it for classifying clusters.

[8]End-to-end latency largely depends on the distance between the source and destination and thus we use last hop latency.

[9]We use last hop latency and variance for differentiating fiber from DSL and cable, respectively, as we will describe in Section 3.2.2.

[10]Although Bitnodes enables us to identify fixed broadband clients, it does not provide information about connection types.

better criterion for distinguishing between cable and fiber clients from the perspective of information gain. We thus use the variance of RTTs for classifying the clusters of ISPs that provide both fiber and cable services, and use the last hop latency for ISPs with fiber and DSL services. We determine the thresholds for classification by using the method described in Section 3.1.2, i.e., we find threshold values that maximize information gain. As a result, we classify the clusters of ISPs providing fiber and DSL services into fiber, DSL or both fiber and DSL if at least 93% of the addresses in the cluster have the last hop latency less than 8.88ms, at most 31% have the latency less than 8.88, or otherwise. For ISPs with fiber and cable services, we use the variance of RTTs 1.483, and 83% or 24% for classifying clusters as fiber or cable.

4 MEASUREMENT RESULTS

We apply the name clustering and media type inference techniques to 103 broadband ISPs[11] that we identify from an international broadband pricing dataset made available by Google [11]. We identify rDNS names for client addresses and analyze whether the keywords in rDNS names are intuitive. We also investigate whether ISPs use different keywords in rDNS names for clients with different connection types.

We found dominant rDNS names for 91 ISPs. However, the keywords in rDNS names of 27 ISPs are being used for both router and client addresses. Some of these keywords indicate client addresses. For example, an Italian ISP called Tiscali uses the keyword "client" for many routers as well as client addresses. Kabel Deutschland (a German ISP) and TDC (a Danish ISP) frequently use the keywords "dynamic" and "customer" for their routers. This may imply that the ISPs did not update the rDNS names of addresses that were reallocated from clients to routers.

64 ISPs have distinct rDNS names for client addresses. As expected, the ISPs have their own naming conventions and the keywords in rDNS names are very diverse. 32 ISPs use intuitive keywords including customer, client, pool, dynamic, ppp, dsl and ftth. The others have non-intuitive or non-English keywords. For example, a Finnish ISP Elisa and a Swedish ISP Com Hem use the keywords "laajakaista" and "bredband", that both mean "broadband". A British ISP TalkTalk has the keyword "AS43234" and several ISPs simply use IP addresses followed by ISP names as rDNS names.

We next analyze connection types of client addresses having common rDNS keywords. Of the 64 ISPs that have distinct rDNS names for client addresses, 29 ISPs use common keywords for clients with different connection types. While most of the ISPs use common keywords that generally indicate client addresses such as dynamic, ppp and client, there are ISPs that use common keywords that indicate specific connection types. For example, several ISPs including Bell Canada, Mweb, Viettel and Webafrica use the keyword "dsl" for both DSL and fiber clients. Shaw communications includes the string "cable" in the rDNS names that are used for both cable and fiber clients. No ISP uses keywords indicating fiber services (e.g., ftth) for non-fiber connections. Considering that ISPs upgrade access networks from DSL/cable to fiber but not vice versa, these results suggest that some ISPs do not keep rDNS names up-to-date.

To summarize, rDNS names need to be carefully used for identifying and characterizing broadband client addresses in that many ISPs use common keywords in rDNS names for clients with different connection types and even for clients and routers. Even if ISPs have different rDNS names for different types of hosts, the keywords in names can be non-intuitive or even be counter-intuitive. Our technique resolves these issues. It can detect whether ISPs uses informative rDNS names and can extract information even from names with non-intuitive keywords. It can also avoid misclassification of rDNS names with counter-intuitive keywords. For proper use of rDNS names in research on broadband Internet clients, we make a list of the identified keywords publicly available at http://www.cs.umd.edu/~ydlee/rdns.

[11] We map ISP names to IPv4 addresses using the Maxmind GeoLite ASN databases and cross-check using the CAIDA's AS to organization mapping dataset (Section 2.1.2). By cross-checking, we could prevent mis-classification of AS327939, AS20668, AS44398, AS19244, AS45898 and AS45899 as ASNs of broadband ISPs.

5 RELATED WORK

5.1 Inferring properties from rDNS names

Many ISPs encode properties of hosts in their rDNS names such as IP type (dynamic or static), geographic locations and subscription type (business or residential). This enabled researchers to infer router locations [1] and whether addresses are dynamically allocated [30] from rDNS names.

More recently, approaches that use machine learning techniques to extract geographic locations [3] and discover the structure of DNS names [2] have been proposed. Huffaker et al. [3] identified valid rules that map rDNS names to geographical locations using a decision tree algorithm, and built general rules by combining related rules. Whereas they used decision tree learning which is supervised learning, we use clustering that does not require training data, which is challenging to obtain in many cases.

Chabarek et al. [2] utilized clustering to identify naming conventions. They applied clustering to feature vectors generated by parsing rDNS names. This approach is similar to our name clustering approach, however, it did not find naming conventions for broadband client hosts because the feature vector used in the clustering did not contain a feature for client hosts. It is possible to add features, but it requires finding keywords for client hosts in rDNS names (and placing them in the dictionary or developing regular expressions that match the keywords). The problem is that finding keywords is a very challenging task because ISPs use their own keywords many of which are non-intuitive. Our name clustering technique does not suffer from this problem. It only requires information that can be easily obtained.

5.2 Detecting broadband client addresses

Using P2P crawlers The fact that hosts are participating in P2P systems is good evidence that they are end hosts. If those hosts have rDNS names containing broadband ISP names, it indicates that they are end hosts of broadband access networks. The detection of broadband client addresses in [4, 31] was performed by this idea. Haeberlen et al. [4] used a list of Napster and Gnutella peers detected in a previous study [32], and Siekkinin et al. [31] used the KAD crawler [33] that finds hosts participating in Kademlia. A problem of using P2P crawlers is the potential bias in the selected addresses, in that people using P2P systems may have different types of broadband services to support their P2P applications, or may keep the link more saturated than most broadband access links are. Broadband client detection using rDNS names can avoid this bias.

Using reverse DNS names Dischinger et al. [5] used rDNS names to select broadband client addresses. They found patterns in rDNS names and identified IP address ranges of client hosts based on the patterns. For example, they found that the pattern for client hosts of BellSouth is adsl-*.bellsouth.net. ThunderPing [6] also used the patterns in rDNS names. It identified /24 blocks that may contain broadband client addresses by resolving IP addresses ending in .1, .44 and .133 to rDNS names, and checking if any of their suffixes matches well-known broadband ISPs. From those /24 blocks, it selected the addresses with rDNS names matching patterns as client addresses. Although these approaches are similar to ours in using patterns in rDNS names, they used hand-crafted patterns.

6 CONCLUSIONS

The results in this paper suggest that rDNS names have information about host type (i.e., whether hosts are clients or routers) and connection type (DSL, cable, fiber or mobile). However, decoding rDNS names is a challenging task that cannot be achieved by the existing approach that uses intuitive keywords. The main obstacle is that many ISPs use non-intuitive keywords to encode information about their hosts. The approach based on intuitive keywords can also lead to misinterpretation of rDNS names in that rDNS names of several ISPs often contain counter-intuitive keywords due to staleness of rDNS names. The clustering-based name classifier technique developed in this paper addresses these issues because it does not require any pre-defined keyword. The rDNS name patterns identified by our methodology are publicly available (http://www.cs.umd.edu/~ydlee/rdns) and can help to improve the reliability of research that uses rDNS names for characterizing edge networks.

This paper has focused on discovering host and connection types by using rDNS names. In future work, we plan to study the extraction of other information in rDNS names such as address allocation type and geographic locations. In particular, we will investigate whether router geolocation rDNS rule sets such as DRoP [3] and undns [1] can be used for geolocation of end hosts.

Another direction of future work is to extend our methodology that distinguishes between rDNS names for different connection types so that it can apply to addresses with non-informative or no rDNS names. Many ISPs use common rDNS name patterns for clients with different connection types. There are also ISPs that do not maintain rDNS names. We plan to adapt our methodology to the sets of or individual addresses having no rDNS names. This will help to increase the number of characterizable hosts.

ACKNOWLEDGMENTS

We thank our shepherd, Alberto Dainotti, and the anonymous reviewers for their helpful comments. This research was supported in part by NSF grant CNS-1619048.

REFERENCES

[1] N. Spring, R. Mahajan, and D. Wetherall, "Measuring ISP topologies with rocketfuel," in *SIGCOMM*, 2002.
[2] J. Chabarek and P. Barford, "What's in a name?: decoding router interface names," in *Proc. the 5th ACM workshop on HotPlanet*, 2013.
[3] B. Huffaker, M. Fomenkov, *et al.*, "Drop: DNS-based router positioning," *ACM SIGCOMM Computer Communication Review*, vol. 44, no. 3, pp. 5–13, 2014.
[4] A. Haeberlen, M. Dischinger, K. P. Gummadi, and S. Saroiu, "Monarch: a tool to emulate transport protocol flowsover the Internet at large," in *IMC*, 2006.
[5] M. Dischinger, A. Haeberlen, K. P. Gummadi, and S. Saroiu, "Characterizing residential broadband networks," in *IMC*, 2007.
[6] A. Schulman and N. Spring, "Pingin' in the rain," in *IMC*, 2011.
[7] L. Quan, J. Heidemann, and Y. Pradkin, "When the internet sleeps: correlating diurnal networks with external factors," in *IMC*, 2014.
[8] X. Cai and J. Heidemann, "Understanding block-level address usage in the visible internet," in *SIGCOMM*, 2010.
[9] M. Sullivan and L. Munoz, "RFC Draft: Suggested generic DNS naming schemes for large networks and unassigned hosts," Oct. 2006.
[10] "Project Sonar reverse DNS dataset." https://scans.io/study/sonar.rdns.
[11] Policy by the numbers, "International broadband pricing study: Updated dataset." https://policybythenumbers.googleblog.com/2015/12/global-broadband-pricing-study-updated.html.
[12] ITU, "The key ICT data." http://www.itu.int/en/ITU-D/Statistics/Pages/stat/.
[13] MaxMind, Inc., "Geolite ASN." http://dev.maxmind.com/geoip/legacy/geolite/.
[14] Y. Lee and N. Spring, "Identifying and aggregating homogeneous IPv4 /24 blocks with Hobbit," in *IMC*, 2016.
[15] CAIDA, "The CAIDA UCSD IPv4 Routed /24 Topology Dataset." http://www.caida.org/data/active/ipv4_routed_24_topology_dataset.xml.
[16] S. C. Johnson, "Hierarchical clustering schemes," *Psychometrika*, vol. 32, no. 3, pp. 241–254, 1967.
[17] G. Navarro, "A guided tour to approximate string matching," *ACM computing surveys (CSUR)*, vol. 33, no. 1, pp. 31–88, 2001.
[18] R Documentation, "Hierarchical Clustering." https://stat.ethz.ch/R-manual/R-devel/library/stats/html/hclust.html.
[19] CAIDA, "AS to Organization Mapping Dataset." http://www.caida.org/data/as_organizations.xml.
[20] G. Maier, A. Feldmann, V. Paxson, and M. Allman, "On dominant characteristics of residential broadband Internet traffic," in *IMC*, 2009.
[21] Yeow, Addy, "Bitnodes." https://getaddr.bitnodes.io/.
[22] Time Warner Cable LLC., "Customer network space and DNS patterns." http://postmaster.rr.com/customer_network_space.
[23] R. Padmanabhan, P. Owen, A. Schulman, and N. Spring, "Timeouts: Beware surprisingly high delay," in *IMC*, 2015.
[24] Q. Xu, J. Huang, Z. Wang, F. Qian, A. Gerber, and Z. M. Mao, "Cellular data network infrastructure characterization and implication on mobile content placement," in *SIGMETRICS*, 2011.
[25] K. Zarifis, T. Flach, S. Nori, D. Choffnes, R. Govindan, E. Katz-Bassett, Z. M. Mao, and M. Welsh, "Diagnosing path inflation of mobile client traffic," in *PAM*, 2014.
[26] L. Rokach and O. Maimon, *Data mining with decision trees: theory and applications*. World scientific, 2014.
[27] I. Canadi, P. Barford, and J. Sommers, "Revisiting broadband performance," in *IMC*, 2012.
[28] S. Sundaresan, W. De Donato, N. Feamster, R. Teixeira, S. Crawford, and A. Pescapè, "Broadband Internet performance: a view from the gateway," in *SIGCOMM*, 2011.
[29] RIPE NCC, "Atlas Probes." https://atlas.ripe.net/probes/.
[30] Y. Xie, F. Yu, K. Achan, E. Gillum, M. Goldszmidt, and T. Wobber, "How dynamic are IP addresses?," in *SIGCOMM*, 2007.
[31] M. Siekkinen, D. Collange, G. Urvoy-Keller, and E. W. Biersack, "Performance limitations of ADSL users: A case study," in *PAM*, 2007.
[32] S. Saroiu, P. K. Gummadi, and S. D. Gribble, "A measurement study of peer-to-peer file sharing systems," in *Proc. the Multimedia Computing and Networking (MMCN)*, 2002.
[33] M. Steiner, T. En-Najjary, and E. W. Biersack, "A global view of KAD," in *IMC*, 2007.